

# DIABETIC RETINOPATHY SCREENING REPORT

SIH 2026 | Problem Statement 26038 | MathWorks AI Pipeline | Explainable AI

ID: sample\_fundus      Date: 09-Sep-2026 07:56      Quality: BORDERLINE (2/3)

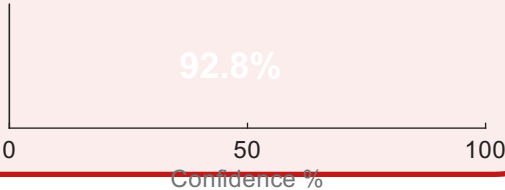
## 1. AI DIAGNOSIS

### PREDICTED DIAGNOSIS

Moderate NPDR

ICDR Severity Level 2 / 4 | AI Confidence 92.8%

URGENT REFERRAL — Ophthalmologist tele-consult within 7 days



## 2. IMAGE QUALITY METRICS

|                                |      |        |
|--------------------------------|------|--------|
| Sharpness (Laplacian Variance) | 19.3 | ✗ FAIL |
| Brightness (Mean Intensity)    | 51.3 | ✓ PASS |
| Contrast (Standard Deviation)  | 31.3 | ✓ PASS |

## 3. CLINICAL LESION EVIDENCE

### Microaneurysms (MA)

|                                                                     |       |
|---------------------------------------------------------------------|-------|
| 24                                                                  | found |
| Earliest DR sign — small vascular microleaks in retinal capillaries |       |

### Hard Exudates (EX)

|                                                                      |       |
|----------------------------------------------------------------------|-------|
| 1                                                                    | found |
| Bright lipid deposits — breakdown of blood-retinal barrier integrity |       |

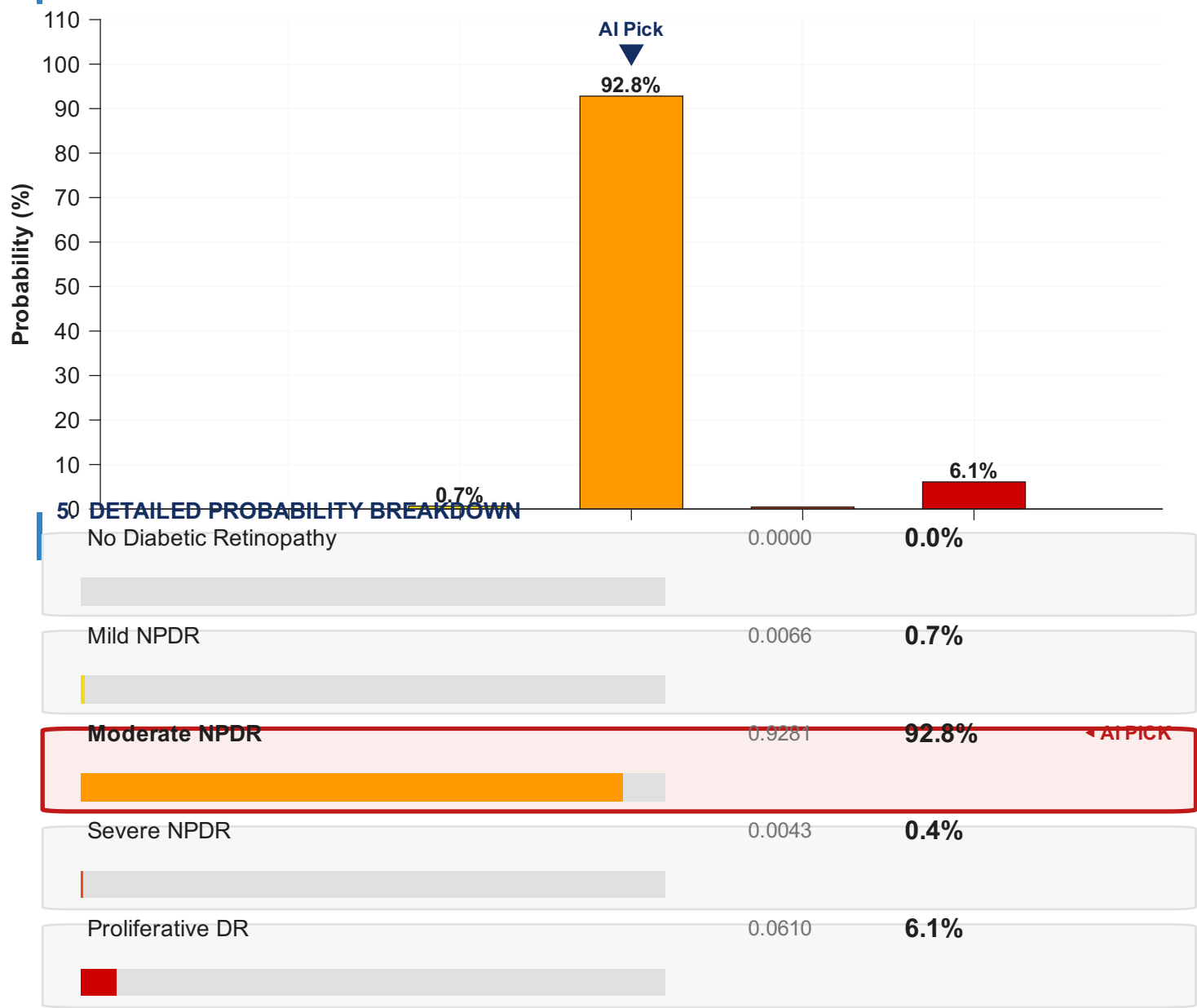
### Hemorrhages (HE)

|                                                          |       |
|----------------------------------------------------------|-------|
| 0                                                        | found |
| Dark blotches — deeper rupture of retinal vascular walls |       |

Optic Disc Centre [X:371 Y:239]    Fovea Centre [X:454 Y:244]

# CLASS PROBABILITY ANALYSIS

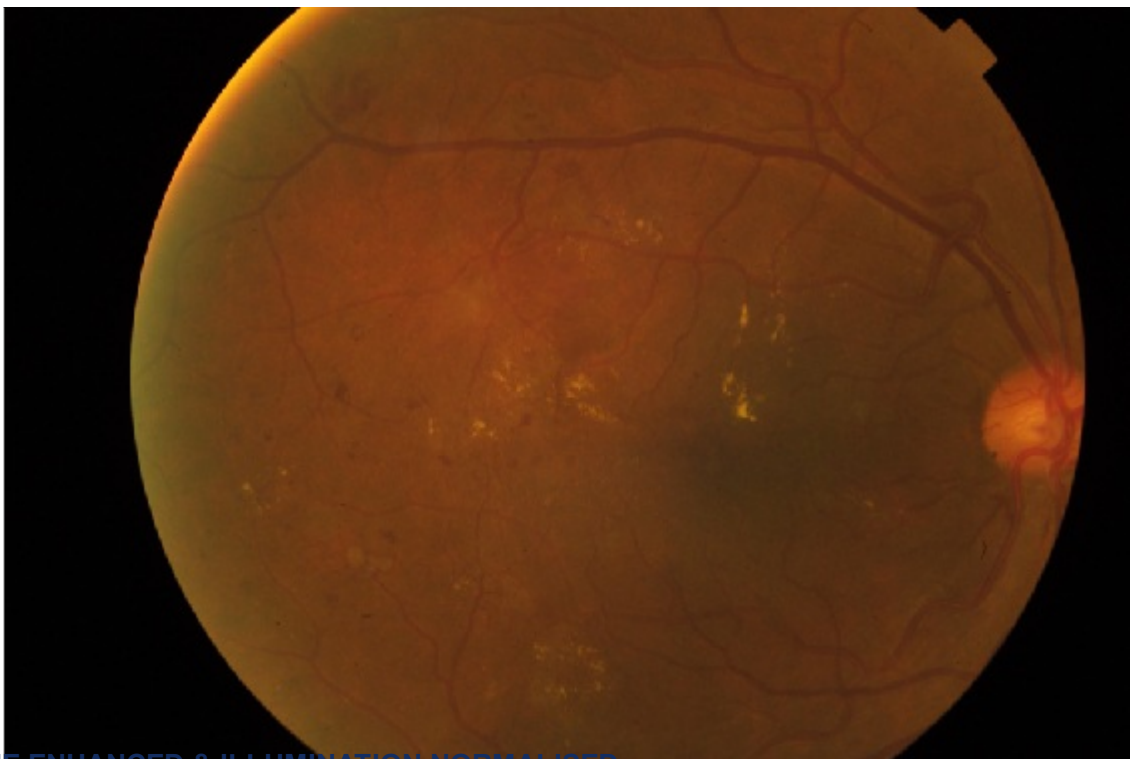
## 4. PROBABILITY DISTRIBUTION ACROSS DR SEVERITY CLASSES



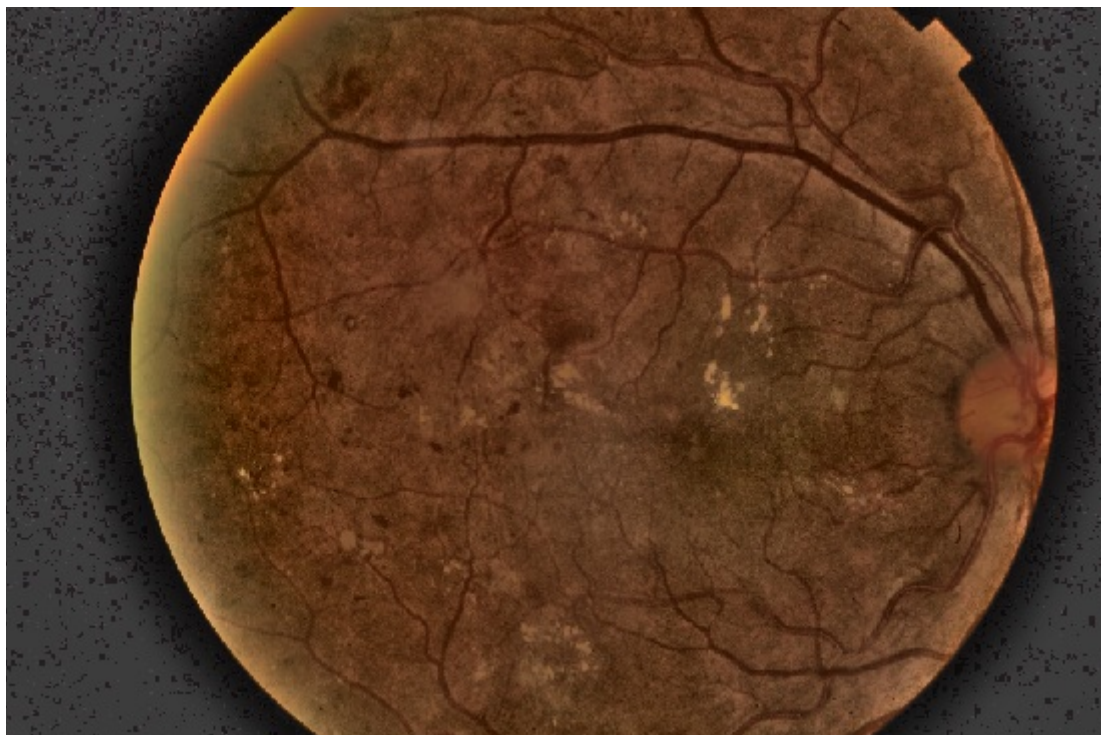
# FUNDUS IMAGE QUALITY & ENHANCEMENT

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## A. ORIGINAL FUNDUS IMAGE



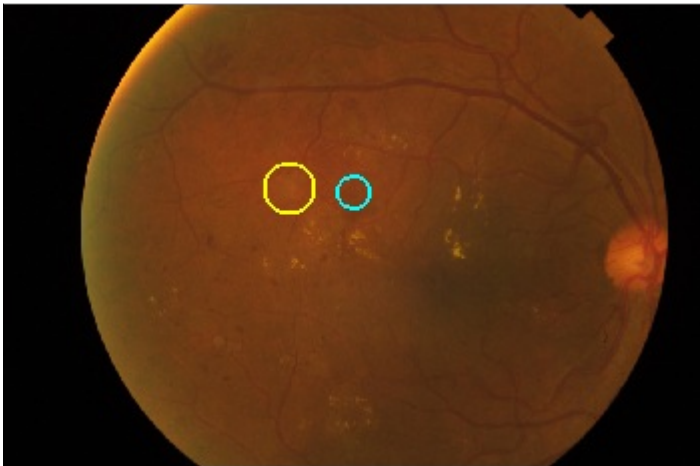
## B. CLAHE ENHANCED & ILLUMINATION NORMALISED



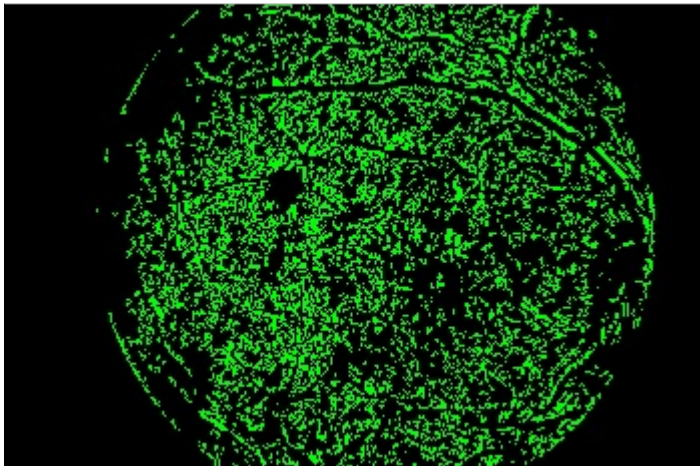
# RETINAL SEGMENTATION & EXPLAINABLE AI

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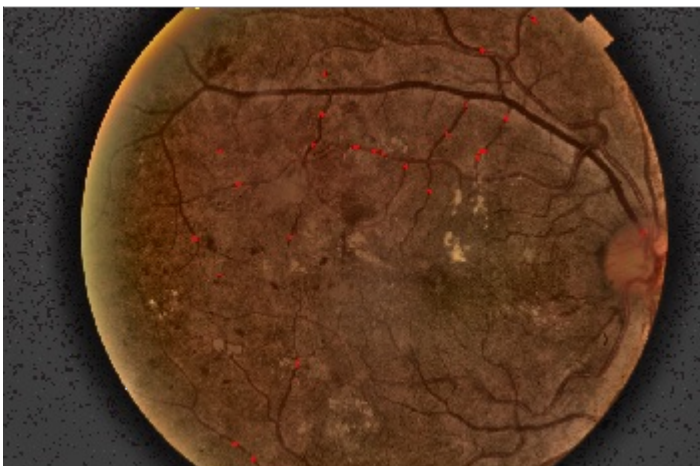
A. Anatomical Landmarks [OD = Yellow | Fovea = Cyan]



B. Vessel Network [Top-Hat Morphological Segmentation]



C. Lesion Map [MA=24 Red | EX=1 Yellow | HE=0 Magenta]



D. Grad-CAM Explainability [Moderate NPDR | 92.8% Confidence]

